

Number-Theoretic Structures:

Phillips Framework and FFGFT — Systematic Comparison

(revised edition — σ -orphan finding added)

Working document. Paul Phillips submitted a number-theoretic framework to the IPI group (23 files, SHA-256 pinned). Bundle not yet fully evaluated. All central claims Python-verified. **Central finding:** The σ -orphan primes in the 13-smooth ambient are exactly $\{2, 5, 11\}$ — and 5, 11 are exactly the middle terms of the cubic sequence $3 \rightarrow 5 \rightarrow 11 \rightarrow 1321$.

1. Elements of the Phillips Framework

1.1 σ -Function and $h=15$

$\sigma(n) = \sum_{d|n} d$. For $n=15$:

$$\sigma(15) = (1+3)(1+5) = 24 = 4! \quad \sigma(14) = (1+2)(1+7) = 24 \quad [\sigma(14)=\sigma(15)] \quad \checkmark$$

$h=15$ is a contact point of two σ -branches.

1.2 σ -Orphans and L_0 -Skeleton

A σ -orphan is a number that never appears as $\sigma(n)$ for any n .

Python finding (central result): The σ -orphan primes in the 13-smooth ambient $\langle 2,3,5,7,11,13 \rangle$ are exactly **$\{2, 5, 11\}$** — three elements only. 3 is not an orphan: $\sigma(2)=3 \checkmark$. 7: $\sigma(4)=7 \checkmark$. 13: $\sigma(9)=13 \checkmark$.

1.3 Cubic Recursion $3 \rightarrow 5 \rightarrow 11 \rightarrow 1321$

All terms prime. Formula in bundle (not yet extracted).

$$1321 \text{ prime } \checkmark \cdot 1321 = 2^3 \times 3 \times 5 \times 11 + 1 = 1320 + 1 \quad \checkmark \quad 1320 = 8 \times 3 \times 5 \times 11 \text{ — product of all predecessors } \times 2^3$$

1.4 13-Smooth Numbers · V_4/CPT · Epitaxy/Homology

Ambient $\langle 2,3,5,7,11,13 \rangle$. First outside prime: 17. $V_4 = \mathbb{Z}_2 \times \mathbb{Z}_2 = \{e, C, P, T\}$ as natural symmetry group of σ -arithmetic.

2. FFGFT Parallels

2.1 Structural Primes $\{2,3,5\}$ and 13-Smooth Ambient

$$\xi = 4/30000 = 1/7500 = 1/(3 \times 2^2 \times 5^4) \quad [\xi^{-1} = 7500 \quad \checkmark]$$

Prime	Power in ξ^{-1}	Geometric meaning in FFGFT
2	$2^2 = 4$	Spacetime dimensions T^4

Prime	Power in ξ^{-1}	Geometric meaning in FFGFT
3	3^1	Spatial dimensions
5	5^4	Fractal layer structure
7	$\kappa=7$ cosmological	internally constrained
11	lepton sector exponent	recursion
13	$\alpha^{-1}=(\xi\cdot\varphi^3\cdot 13)^{-1}=136.19$	Method 1 ✓

17 (first outside prime) has no structural role in FFGFT. Consistent.

2.2 Central Finding: σ -Orphans and the Cubic Sequence

Strongest result — Python-verified: σ -orphan primes in 13-smooth ambient: {2, 5, 11} Cubic sequence: $3 \rightarrow \mathbf{5} \rightarrow \mathbf{11} \rightarrow 1321$ The terms 5 and 11 are exactly the σ -orphan primes (other than 2). The starting value 3 is not an orphan ($\sigma(2)=3$). The recursion begins at a σ -value and then runs exclusively through σ -orphan primes from the 13-smooth ambient. **This is not a numerical coincidence.**

In FFGFT, 5 and 11 appear in the lepton sector: $r_\varphi = 25/9 = 5^2/3^2$; recursion depth $N\approx 8.4$ contains factor 5 ($N\approx 25/3$).

2.3 $h=15$ as {3,5}-Core of 30000

$$30000 = 2^4 \times 3 \times 5^4 = 2^4 \times 5^3 \times 15 = 2000 \times 15 \checkmark$$

$$\sigma(15) = 24 = 4! = |S_4| \text{ and } T^4 \text{ has permutation symmetry } S_4 \text{ (order 24)}$$

2.4 V_4 /CPT — Genuine Structural Overlap

FFGFT derives V_4 from T^4 topology: $CPT \subset \mathbb{Z}_2^4$ as the subgroup preserving Lorentz signature. Phillips' σ -arithmetic generates the same group — neither postulated, both derived.

3. Overall Assessment

Overlap	Strength	Status
{2,3,5} as ξ -core	strong	derived from T^4 ✓
7, 11, 13 in ambient	strong	all in FFGFT ✓
17 as first outside prime	consistent	no FFGFT role ✓
σ -orphans {2,5,11} = seq. $3\rightarrow 5\rightarrow 11$	★ strong	Python ✓ — new

Overlap	Strength	Status
$h=15 = 3 \times 5$ as ξ -core	notable	non-trivial
$\sigma(15) = 24 = S_4 $	interesting	T^4 symmetry
V_4/CPT	genuine overlap	both derived ✓
Recursion depth $N \approx 8$	notable	mechanism open
$1321 = 2^3 \cdot 3 \cdot 5 \cdot 11 + 1$	interesting	prime ✓
E_8 connection	unknown	bundle pending
Cubic formula	unknown	bundle pending

Conclusion: The overlaps in structural primes, V_4/CPT , the 13-smooth ambient, and in particular the identity $\{\sigma\text{-orphan primes}\} \cap \{\text{cubic sequence}\} = \{5, 11\}$ are too specific to be purely coincidental. The bundle must be fully evaluated — especially the cubic recursion formula, the E_8 section, and the exact construction of the L_0 -skeleton.

Next Steps

- (1) Read full bundle: cubic formula, E_8 section
- (2) Question to Paul: how exactly is the 8-level L_0 -skeleton constructed?
- (3) Why does the sequence start at 3 (not a σ -orphan)?
- (4) Does Phillips' framework yield ξ or an approximation?